

Sales Prediction Using Python

Submitted By

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Project Title: Sales Prediction
Using Python

1. Abstract

This project aims to predict product sales based on advertising expenditure on TV, Radio, and Newspaper using the Linear Regression algorithm. The model analyzes historical advertising data to estimate future sales, helping businesses make informed marketing decisions. The project includes data preprocessing, exploratory data analysis (EDA), model training, evaluation, and business insights.

2. Introduction

Sales prediction is an important application of machine learning in business. Companies invest heavily in advertising, and understanding how different advertising channels influence sales helps optimize marketing budgets.

In this project, a Linear Regression model is developed to predict sales using advertising expenditure on:

- TV

- Radio
- Newspaper

3. Objectives

- Predict product sales using machine learning.
 - Analyze the impact of advertising channels.
 - Perform data cleaning and preprocessing.
 - Build a Linear Regression model.
 - Evaluate the model's performance.
 - Provide business recommendations.
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4. Tools and Technologies

Tool

Purpose

Python

Programming Language

Jupyter Notebook Development Environment

Pandas

Data Manipulation

Tool Purpose

NumPy	Numerical Computation
Matplotlib	Data Visualization
Seaborn	Statistical Visualization
Scikit-learn	Machine Learning

5. Dataset Description

The dataset contains **200 records** and **4 features**.

Feature	Description
TV	TV advertising budget
Radio	Radio advertising budget
Newspaper	Newspaper advertising budget
Sales	Product sales

6. Methodology

The project follows these steps:

- 1.Import Libraries
- 2.Load Dataset
- 3.Data Cleaning
- 4.Exploratory Data Analysis (EDA)
- 5.Correlation Analysis
- 6.Feature Selection
- 7.Train-Test Split
- 8.Train Linear Regression Model
- 9.Predict Sales
10. Evaluate Model
11. Generate Business Insights

7. Exploratory Data Analysis (EDA)

During EDA, the following visualizations were created:

- Correlation Heatmap
- Pairplot
- Histograms
- Boxplots
- Scatter Plots

Key Findings:

- TV advertising has the strongest positive correlation with Sales.
- Radio advertising has a moderate positive correlation.
- Newspaper advertising has the weakest correlation.

8. Machine Learning Model

The **Linear Regression** algorithm was used to predict sales.

Input Features:

- TV
- Radio
- Newspaper

Target Variable:

- Sales

9. Model Performance

Metric Value

MAE **1.4608**

MSE **3.1741**

RMSE **1.7816**

R² Score **0.8994**

The model explains approximately **90%** of the variation in sales, indicating excellent predictive performance.

10. Business Insights

- TV advertising has the highest impact on product sales.
 - Radio advertising also contributes significantly.
 - Newspaper advertising has the least impact.
 - Businesses should prioritize TV and Radio advertising for better returns.
 - The model can help estimate future sales before launching marketing campaigns.
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11. Conclusion

The Sales Prediction model was successfully developed using Linear Regression. The model achieved an R^2 Score of **0.8994**, indicating strong predictive capability. The analysis demonstrates that advertising expenditure is an important factor influencing sales, especially TV and Radio advertising.

12. Future Scope

- Develop a Streamlit web application.
 - Compare Linear Regression with Random Forest and XGBoost.
 - Deploy the model online.
 - Use larger real-world datasets.
 - Add hyperparameter tuning for improved performance.
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13. References

1. Python Documentation
 2. Pandas Documentation
 3. Scikit-learn Documentation
 4. Matplotlib Documentation
 5. Seaborn Documentation
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14. Appendix

Include:

- Sample dataset
- Model output
- Prediction example
- Screenshots of the notebook
- Important graphs